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Display with sliding display to expose a camera

Abstract

A display module and a display device are provided. The display module includes a fixed function portion and a sliding display portion disposed oppositely. The fixed function portion includes a glass cover plate and a function layer disposed between the glass cover plate and the sliding display portion. The sliding display portion includes a first sub-part close to a side of the fixed function portion and a second sub-part bent and extended from the first sub-part and away from the side of the fixed function portion. The sliding display portion may slide relative to the fixed function portion.

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Background/Summary

RELATED APPLICATIONS
(1) This application is a Notional Phase of PCT Patent Application No. PCT/CN2021/116606 having international filing date of Sep. 6, 2021, which claims the benefit of priority of Chinese Patent Application No. 202110668051.1 filed on Jun. 16, 2021. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.
FIELD OF INVENTION
(2) The present disclosure relates to the technical field of display, and particularly to a display module and a display device comprising the display module.
BACKGROUND
(3) A full screen is a large-screen smart phone solution proposed by mobile phone manufacturers in recent years. In order to make a mobile phone have a higher screen ratio and better visual effects, there are various screen designs such as “notch screen”, “water drop screen”, and “pop-up Camera”. The above solutions have disadvantages of high cost and poor practicability. Furthermore, there is still a camera installation position on a main screen, which cannot be regarded as a full screen in a true sense, and cannot meet users demand for a full screen. A “full screen” that has a higher screen ratio and is more concise and easy-to-use is what users are looking for, which can bring users a better visual enjoyment and use experience.
(4) A flexible display screen is made of soft materials and is a bendable, rollable, and foldable display device. The flexible display screen has characteristics of lighter, thinner, and longer durability. The flexible display screen is a display screen device that can be moved like a scroll. Currently, a camera assembly is usually disposed under the flexible display screen. When there is no need to take a picture, the flexible display screen covers the camera assembly for full-screen display. When there is a need to take a picture, a part of the flexible display screen is rolled and accommodated to expose the camera assembly for taking pictures.
(5) However, in a current laminated design of the flexible display screen, laminated layers participating in rolling up and accommodating are relatively thick. Therefore, when the flexible display screen is bent, the layers are easily peeled off and cracked, resulting in moisture intrusion into the flexible display screen during a test, and subsequent problems of poor display.
SUMMARY OF DISCLOSURE
(6) The present disclosure provides a display module and a display device, which can reduce a thickness of a laminated structure participating in rolling up in the display module, thereby improving a rollability and reliability of the display module.

(7) The present disclosure provides a display module comprising: a fixed function portion; and a sliding display portion disposed opposite to the fixed function portion and comprising a first sub-part close to a side of the fixed function portion and a second sub-part bent and extended from the first sub-part and away from the side of the fixed function portion; wherein the fixed function portion comprises a glass cover plate and a function layer disposed between the glass cover plate and the sliding display portion; the sliding display portion slides relative to the fixed function portion, so that the display module is switched between a first state and a second state; in the first state, the first sub-part is located on a side of the sliding display portion close to the fixed function portion; and in the second state, at least part of the first sub-part slides to a side of the sliding display portion away from the fixed function portion.
(8) In an embodiment, the sliding display portion further comprises a display panel. A side of the display panel close to the function layer is a first sliding surface, and a side of the function layer close to the display panel is a second sliding surface.
(9) In an embodiment, the function layer comprises a polarizing layer, and the polarizing layer is disposed on a side of the glass cover plate close to the display panel.
(10) In an embodiment, the function layer further comprises an ultra-thin glass layer, the ultra-thin glass layer is disposed on a side of the polarizing layer close to the display panel, and a side of the ultra-thin glass layer close to the display panel is the second sliding surface.
(11) In an embodiment, the function layer further comprises a touch layer, and the touch layer is disposed between the glass cover plate and the polarizing layer.
(12) In an embodiment, the first sliding surface and the second sliding surface are provided with a first hardened coating, and a surface contact angle of the first hardened coating is greater than or equal to 100°.
(13) In an embodiment, the sliding display portion further comprises a first support assembly disposed on a side of the display panel away from the fixed function portion. The first support assembly comprises a support layer. The support layer comprises a rigid part and a flexible part whose one end is connected to the rigid part and the other end is bent and away from the side of the fixed function portion. The flexible part comprises a plurality of openings, and a distribution density of the openings on a side of the flexible part close to the rigid part is less than a distribution density of the openings on a side of the flexible part away from the rigid part.
(14) In an embodiment, the first support assembly further comprises a buffer film disposed between the support layer and the sliding display portion, and the buffer film is made of a polyurethane elastomer material.
(15) In an embodiment, the display module further comprises a second support assembly disposed on the side of the sliding display portion away from the fixed function portion. The sliding display portion slides relative to the second support assembly.
(16) In an embodiment, a side of the sliding display portion close to the second support assembly is a third sliding surface, and a side of the second support assembly close to the sliding display portion is a fourth sliding surface. The third sliding surface and the fourth sliding surface are provided with a second hardened coating, and a surface contact angle of the second hardened coating is greater than or equal to 100°.
(17) In an embodiment, the display module further comprises a camera assembly disposed on the side of the sliding display portion away from the fixed function portion. When the display module is in the first state, an orthographic projection of the camera assembly on the fixed function portion is located within an orthographic projection of the sliding display portion on the fixed function portion. When the display module is in the second state, the orthographic projection of the camera assembly on the fixed function portion is outside the orthographic projection of the sliding display portion on the fixed function portion.
(18) The present disclosure further provides a display device comprising a display module. The display module comprises: a fixed function portion; and a sliding display portion disposed opposite

to the fixed function portion and comprising a first sub-part close to a side of the fixed function portion and a second sub-part bent and extended from the first sub-part and away from the side of the fixed function portion; wherein the fixed function portion comprises a glass cover plate and a function layer disposed between the glass cover plate and the sliding display portion; the sliding display portion slides relative to the fixed function portion, so that the display module is switched between a first state and a second state; in the first state, the first sub-part is located on a side of the sliding display portion close to the fixed function portion; and in the second state, at least part of the first sub-part slides to a side of the sliding display portion away from the fixed function portion.
(19) In an embodiment, the sliding display portion further comprises a display panel. A side of the display panel close to the function layer is a first sliding surface, and a side of the function layer close to the display panel is a second sliding surface. The first sliding surface and the second sliding surface are provided with a first hardened coating, and a surface contact angle of the first hardened coating is greater than or equal to 100°.
(20) In an embodiment, the function layer comprises a polarizing layer, and the polarizing layer is disposed on a side of the glass cover plate close to the display panel.
(21) In an embodiment, the function layer further comprises an ultra-thin glass layer, the ultra-thin glass layer is disposed on a side of the polarizing layer close to the display panel, and a side of the ultra-thin glass layer close to the display panel is the second sliding surface.
(22) In an embodiment, the function layer further comprises a touch layer, and the touch layer is disposed between the glass cover plate and the polarizing layer.
(23) In an embodiment, the display module further comprises a camera assembly disposed on the side of the sliding display portion away from the fixed function portion. When the display module is in the first state, an orthographic projection of the camera assembly on the fixed function portion is located within an orthographic projection of the sliding display portion on the fixed function portion. When the display module is in the second state, the orthographic projection of the camera assembly on the fixed function portion is outside the orthographic projection of the sliding display portion on the fixed function portion.
(24) In an embodiment, the display device further comprises a middle frame. The fixed function portion is fixedly connected to the middle frame, and the sliding display portion is movably connected to the middle frame, so that the display module is switched between the first state and the second state.
(25) In an embodiment, the middle frame is provided with a sliding assembly. The sliding assembly comprises a sliding rail fixedly connected to the middle frame and a sliding block slidably connected to the sliding rail. The sliding display portion is fixedly connected to the sliding block.
(26) In an embodiment, the display device further comprises an elastic member and a driving assembly. One end of the elastic member is connected to the middle frame, and the other end of the elastic member is connected to the first sub-part. The driving assembly is disposed on the middle frame. One end of the driving assembly is connected to the second sub-part, so that the display module is switched between the first state and the second state.
(27) Compared with the prior art, in the present disclosure, a function layer is disposed between a sliding display portion and a glass cover plate, and the sliding display portion can slide relative to the function layer and the glass cover plate to be bent and accommodated. That is, the function layer does not participate in bending, so that a thickness of a laminated structure participating in the bending in the display module is reduced, thereby reducing a strain/dislocation generated in a bending process of the sliding display portion, improving a bendability of the display module, and improving a reliability and service life of the display module.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Specific implementation of the present disclosure will be described in detail below in conjunction with accompanying drawings to make technical solutions and beneficial effects of the present disclosure obvious.

(2) FIG. **1** is a schematic structural diagram of a display panel in the prior art.

(3) FIG. **2** is a schematic diagram of a structure of a display panel according to an embodiment of the present disclosure.

(4) FIG. **3**A is a schematic structural diagram of a first state of the display panel according to an embodiment of the present disclosure.

(5) FIG. **3**B is a schematic structural diagram of a second state of the display panel according to an embodiment of the present disclosure.

(6) FIG. **4** is a schematic structural diagram of a distribution of a plurality of openings according to an embodiment of the present disclosure.

(7) FIG. **5** is a schematic diagram of another structure of the display panel according to an embodiment of the present disclosure.

(8) FIG. **6** is a schematic diagram of another structure of the display panel according to an embodiment of the present disclosure.

(9) FIG. **7** is a schematic diagram of another structure of the display panel according to an embodiment of the present disclosure.

(10) FIG. **8** is a schematic rear view of the display panel according to an embodiment of the present disclosure.

(11) FIG. **9** is an exploded schematic structural diagram of the display panel according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(12) Technical solutions in embodiments of the present disclosure will be clearly and completely described below in conjunction with accompanying drawings in the embodiments of the present disclosure. It is apparent that the described embodiments are merely a part of the embodiments of the present disclosure and not all embodiments. All other embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative labor are within claimed scope of the present disclosure.

(13) The following description provides different embodiments or examples illustrating various structures of the present invention. In order to simplify the description of the present disclosure, only components and settings of specific examples are described below. They are only examples and are not intended to limit the present invention. Furthermore, reference numerals and/or letters may be repeated in different examples of the present disclosure. Such repetitions are for simplicity and clarity, which per se do not indicate relations among the discussed embodiments and/or settings. Furthermore, the present disclosure provides various examples of specific processes and materials, but those skilled in the art can be aware of application of other processes and/or use of other materials.

(14) Currently, an under-screen camera is usually disposed to achieve a full-screen display device. Please refer to FIG. **1**, a display device comprises a panel body **1**, an additional assembly **2** attached to a light-emitting side of the panel body **1**, a backplane support assembly **3** disposed on a side of the panel body **1** away from the additional assembly **2**, a glass cover plate **4** disposed on a side of the additional assembly **2** away from the panel body **1**, and a camera **5** disposed on a side of the backplane support assembly **3** away from the panel body **1**. The camera **5** is disposed under the panel body **1**, thereby not affecting full-screen display of the display device. When functions such as taking pictures are needed, an end of the panel body **1** away from the camera **5** will slide and bend to a side of the panel body **1** away from the glass cover plate **4**, thereby driving an end of the panel body **1** close to the camera **5** to move horizontally and expose the camera **5** to realize

functions such as taking pictures. However, during a bending process of the panel body **1**, the additional assembly **2** and the backplane support assembly **3** will be bent together with the panel body **1**. Because a thickness of a laminated structure participating in bending is larger, layers are easily peeled off and cracked, resulting in risks of moisture and oxygen intrusion and poor display.
(15) Accordingly, the present disclosure provides a display module. Please refer to FIG. **2**, FIG. **3**A, and FIG. **3**B. The display module comprises a fixed function portion **101** and a sliding display portion **102** disposed oppositely.
(16) The fixed function portion **101** comprises a glass cover plate **30** and a function layer **20**. The function layer **20** is disposed between the glass cover plate **30** and the sliding display portion **102**.
(17) The sliding display portion **102** comprises a first sub-par **1021** close to a side of the fixed function portion **101** and a second sub-part **1022** bent and extended from the first sub-part **1021** and away from the side of the fixed function portion **101**.
(18) The sliding display portion **102** slides relative to the fixed function portion **101**, so that the display module is switched between a first state and a second state. In the first state, the first sub-part **1021** is located on a side of the sliding display portion **102** close to the fixed function portion **101**. In the second state, at least part of the first sub-part **1021** slides to a side of the sliding display portion **102** away from the fixed function portion **101**.
(19) In an implementation of the present disclosure, the function layer **20** is disposed between the sliding display portion **102** and the glass cover plate **30**, and the sliding display portion **102** can slide relative to the function layer **20** and the glass cover plate **30** to be bent and accommodated. That is, the function layer **20** does not participate in bending, so that a thickness of a laminated structure participating in the bending in the display module is reduced, thereby reducing a strain/dislocation generated in a bending process of the sliding display portion **102**, improving a bendability of the display module, and improving a reliability and service life of the display module.
(20) Specifically, please continue to refer to FIG. **2**, FIG. **3**A, and FIG. **3**B. The display module comprises the sliding display portion **102**, the fixed function portion **101** disposed on the sliding display portion **102**, and a camera assembly **50** disposed under the sliding display portion **102**.
(21) The camera assembly **50** is located under the sliding display portion **102** and is covered by the sliding display portion **102**, so that full-screen display can be realized. Furthermore, the sliding display portion **102** can slide relative to the fixed function portion **101**. During a sliding process of the sliding display portion **102**, a first end of the sliding display portion **102** away from the camera assembly **50** will bend away from a side of the fixed function portion **101**. A second end of the sliding display portion **102** close to the camera assembly **50** will also slide toward the first end under a drive of the first end to expose the camera assembly **50**, which can realize a photographing function.
(22) That is, the sliding display portion **102** slides relative to the fixed function portion **101**, so that the display module is switched between the first state and the second state. In the first state, the first sub-part **1021** is located on the side of the sliding display portion **102** close to the fixed function portion **101**. In the second state, at least part of the first sub-part **1021** slides to the side of the sliding display portion **102** away from the fixed function portion **101**.
(23) Furthermore, when the display module is in the first state, an orthographic projection of the camera assembly **50** on the fixed function portion **101** is located within an orthographic projection of the sliding display portion **102** on the fixed function portion **101**. When the display module is in the second state, the orthographic projection of the camera assembly **50** on the fixed function portion **101** is outside the orthographic projection of the sliding display portion **102** on the fixed function portion**101**.
(24) The fixed function portion **101** comprises the glass cover plate **30** and the function layer **20**, and the function layer **20** may be disposed on the glass cover plate **30**, so that the sliding display portion **102** slides relative to the glass cover **30** and the function layer **20**.

(25) It should be noted that, in the present disclosure, a surface of the sliding display portion **102** close to the function layer **20** and a surface of the function layer **20** close to the sliding display portion **102** are provided with a hardened coating, and a surface contact angle of the hardened coating is greater than or equal to 100°, thereby improving hardness and friction resistance of the surfaces. This can reduce a frictional force, so that a width of a gap between the sliding display portion **102** and the function layer **20** is close to 0, which improves display effect of the display module.
(26) A structure of the display module provided by the present disclosure will be described in detail below in conjunction with specific embodiments.
(27) In an embodiment, please refer to FIG. **2**, FIG. **3**A, and FIG. **3**B, the display module comprises the fixed function portion **101** and the sliding display portion **102** disposed opposite to each other, and the camera assembly **50** disposed on the side of the sliding display portion **102** away from the fixed function portion **101**.
(28) The sliding display portion **102** comprises the first sub-par **1021** close to the side of the fixed function portion **101** and the second sub-part **1022** bent and extended from the first sub-part **1021** and away from the side of the fixed function portion **101**. The sliding display portion **102** can slide relative to the fixed function portion **101**, so that at least part of the first sub-part **1021** slides away from the side of the fixed function portion **101** to expose the camera assembly **50**, and the display module is in the second state. And then, the sliding display portion **102** can slide relative to the fixed function portion **101**, so that the at least part of the first sub-part **1021** slides back and close to the side of the fixed function portion **101**, the sliding display portion **102** covers the camera assembly **50**, and the display module is in the first state. By controlling the display module to switch between the first state and the second state, the display module is switched between full-screen display and functions such as taking pictures.
(29) In this embodiment, the fixed function portion **101** comprises the glass cover plate **30** and the function layer **20** disposed between the glass cover plate **30** and the sliding display portion **102**. The function layer **20** comprises a touch layer **24** disposed on a side of the glass cover plate **30** close to the sliding display portion **102**, a polarizing layer **22** disposed on a side of the touch layer **24** close to the sliding display portion **102**, and an ultra-thin glass layer **21** disposed on a side of the polarizing layer **22** close to the sliding display portion **102**.
(30) The touch layer **24** is attached to the glass cover plate **30** through a first adhesive layer **25**, and the touch layer **24** is also attached to the polarizing layer **22** through a second adhesive layer **23**.
(31) Optionally, both the first adhesive layer **25** and the second adhesive layer **23** are made of a pressure sensitive adhesive.
(32) The sliding display portion **102** comprises a display panel **10** and a first support assembly **40** disposed on a side of the display panel **10** away from the fixed function portion **101**. That is, in this embodiment, a laminated structure participating in bending and sliding comprises the display panel **10** and the first support assembly **40**.
(33) In this embodiment, a side of the display panel **10** close to the fixed function portion **101** is a first sliding surface. A side of the function layer **20** close to the display panel **10** is a second sliding surface. That is, the second sliding surface is a side of the ultra-thin glass layer **21** close to the display panel **10**. The first sliding surface and the second sliding surface are provided with a first hardened coating, so as to improve hardness and friction resistance of the first sliding surface and the second sliding surface and increase the service life of the display module. Furthermore, a surface contact angle of the first hardened coating is greater than or equal to 100°, which can reduce a frictional force, so that a width of a gap between the display panel **10** and the ultra-thin glass layer **21** is close to 0, thereby improving the display effect of the display module.
(34) The display panel **10** comprises a thin film transistor array substrate **11**, a pixel light-emitting layer **12** disposed on the thin film transistor array substrate **11**, and an encapsulation layer **13** covering the pixel light-emitting layer **12**. The pixel light-emitting layer **12** may be an OLED light-

emitting layer made of organic light-emitting materials, and is connected to thin-film transistor devices in the thin-film transistor array substrate **11** to realize signal transmission and emit light.
(35) The first support assembly **40** comprises a back plate **41** close to a side of the display panel **10**, a buffer film **42** disposed on a side of the back plate **41** away from the display panel **10**, a support layer **43** disposed on a side of the buffer film **42** away from the display panel **10**, and a protective glue layer **44** disposed on a side of the support layer **43** away from the display panel **10**.
(36) Please refer to FIG. **2** and FIG. **4**, the support layer **43** comprises a rigid part **432** and a flexible part **431** whose one end is connected to the rigid part **432** and the other end is bent and away from the side of the fixed function portion **101**. The flexible part **431** comprises a plurality of openings **4311**, and a distribution density of the openings **4311** on a side of the flexible part **431** close to the rigid part **432** is less than a distribution density of the openings **4311** on a side of the flexible part **431** away from the rigid part **432**.
(37) A direction from the rigid part **432** to the flexible part **431** is defined as a first direction X. The distribution density of the openings **4311** on the side of the flexible part **431** close to the rigid part **432** may increase along the first direction X. The distribution density of the openings **4311** on the side of the flexible part **431** away from the rigid part **432** may remain unchanged along the first direction X. That is, in the present disclosure, the openings **4311** with a gradual distribution density are formed on the side of the flexible part **431** close to the rigid part **432**, which prevents stress concentration at an interface between the rigid part **432** and the flexible part **431** due to the openings during a bending process, thereby improving product yield.
(38) Specifically, a distance a between two adjacent openings **4311** in a direction perpendicular to the first direction X is 0.16 to 2.5 mm. A distance b between two adjacent openings **4311** in the first direction X is 0.114 to 0.271 mm. A distance c between central axes of two adjacent openings **4311** in the first direction X is 0.314 to 0.471 mm. A length d of the openings **4311** in the direction perpendicular to the first direction X is 1.9 to 4.24 mm. Parameters of the openings **4311** may be selected according to actual needs, and are not limited herein.
(39) In this embodiment, the buffer film **42** is made of a polyurethane elastomer material. Compared with current foams, the polyurethane elastomer material has excellent cushioning property, toughness, and wear resistance. Therefore, the buffer film **42** may be made thinner to further reduce a thickness of the sliding display portion **102**, thereby improving a bendability of the sliding display portion **102**.
(40) In the above, in this embodiment, the polarizing layer **22** and the touch layer **24** are both disposed in the function layer **20**, and do not participate in bending and sliding of the sliding display portion **102**, thereby reducing the thickness of the sliding display portion **102**. This improves the bendability of the sliding display portion **102**, and reduces a risk of peeling and cracking of layers of the sliding display portion **102** during the bending process. Furthermore, in this embodiment, the side of the ultra-thin glass layer **21** close to the display panel **10** and a side of the display panel **10** close to the ultra-thin glass layer **21** are provided with the first hardened coating, thereby improving friction resistance and hardness of sliding surfaces of the two sides and increasing a service life of the display module.
(41) In another embodiment, please refer to FIG. **5**, this embodiment differs from the first embodiment in laminated structures of a sliding display portion **102** and a function layer **20**.
(42) In this embodiment, the function layer **20** comprises a polarizing layer **22** disposed on a side of a glass cover plate **30** close to a display panel **10**, and an ultra-thin glass layer **21** disposed on a side of a polarizing layer **22** close to the display panel **10**. The polarizing layer **22** is attached to the glass cover plate **30** through a first adhesive layer **25**.
(43) The sliding display portion **102** comprises the display panel **10**, a first support assembly **40** disposed on a side of the display panel **10** away from a fixed function portion **101**, and a touch layer **24** disposed on a side of the display panel **10** close to the fixed function portion **101**.
(44) It should be noted that the touch layer **24** is not limited to a laminated position in this

embodiment. The touch layer **24** may also be disposed inside the display panel **10** to form an in-cell touch display panel, which is not limited herein.
(45) In addition, a surface of the ultra-thin glass layer **21** close to the touch layer **24** and a surface of the touch layer **24** close to the ultra-thin glass layer **21** are provided with a third hardened coating to improve hardness and friction resistance. This can reduce a frictional force, so that a width of a gap between the display panel **10** and the ultra-thin glass layer **21** is close to 0, thereby improving display effect of the display module.
(46) In this embodiment, the polarizing layer **22** is disposed in the function layer **20**, and do not participate in bending and sliding of the sliding display portion **102**, thereby reducing the thickness of the sliding display portion **102**. This improves a bendability of the sliding display portion **102**, and reduces a risk of peeling and cracking of layers of the sliding display portion **102** during the bending process. Furthermore, in this embodiment, the side of the ultra-thin glass layer **21** close to the touch layer **24** and a side of the touch layer **24** close to the ultra-thin glass layer **21** are provided with the third hardened coating, thereby improving friction resistance and hardness of sliding surfaces of the two sides and increasing a service life of the display module.
(47) In another embodiment, please refer to FIG. **6**, this embodiment differs from the first embodiment in laminated structures of a sliding display portion **102** and a function layer **20**.
(48) The function layer **20** comprises a touch layer **24** disposed between a glass cover plate **30** and the sliding display portion **102**, and the touch layer **24** is attached to the glass cover plate **30** through a first adhesive layer **25**.
(49) The touch layer **24** may comprise a glass substrate and a touch electrode layer disposed on the glass substrate, and a side of the glass substrate close to a display panel **10** is provided with a hardened coating.
(50) The sliding display portion **102** comprises the display panel **10**, a first support assembly **40** disposed on a side of the display panel **10** away from a fixed function portion **101**, and a polarizing layer **22** disposed on a side of the display panel **10** close to the fixed function portion **101**.
(51) A surface of the touch layer **24** close to the polarizing layer **22** and a surface of the polarizing layer **22** close to the touch layer **24** are provided with a fourth hardened coating to improve hardness and friction resistance of the surfaces. Furthermore, in this embodiment, a surface contact angle of the fourth hardened coating is greater than or equal to 100°, which can reduce a frictional force, so that a width of a gap between the polarizing layer **22** and the touch layer **24** is close to 0, thereby improving display effect of the display module.
(52) In this embodiment, the touch layer **24** is disposed in the function layer **20**, and do not participate in bending and sliding of the sliding display portion **102**, thereby reducing the thickness of the sliding display portion **102**. This improves a bendability of the sliding display portion **102**, and reduces a risk of peeling and cracking of layers of the sliding display portion **102** during the bending process. Furthermore, in this embodiment, the side of the polarizing layer **22** close to the touch layer **24** and a side of the touch layer **24** close to the polarizing layer **22** are provided with the fourth hardened coating, thereby improving friction resistance and hardness of sliding surfaces of the two sides and increasing a service life of the display module.
(53) In another embodiment, please refer to FIG. **7**, this embodiment differs from the first embodiment in a laminated structure of a sliding display portion **102**.
(54) Specifically, in this embodiment, a fixed function portion **101** comprises a glass cover plate **30** and a function layer **20** disposed between a glass cover plate **30** and the sliding display portion **102**. The function layer **20** comprises a touch layer **24** disposed on a side of the glass cover plate **30** close to the sliding display portion **102**, a polarizing layer **22** disposed on a side of the touch layer **24** close to the sliding display portion **102**, and an ultra-thin glass layer **21** disposed on a side of the polarizing layer **22** close to the sliding display portion **102**.
(55) The touch layer **24** is attached to the glass cover plate **30** through a first adhesive layer **25**, and the touch layer **24** is also attached to the polarizing layer **22** through a second adhesive layer **23**.

(56) Optionally, both the first adhesive layer **25** and the second adhesive layer **23** are made of a pressure sensitive adhesive.
(57) The sliding display portion **102** comprises a display panel **10**. A surface of the ultra-thin glass layer **21** close to the display panel **10** and a surface of the display panel **10** close to the ultra-thin glass layer **21** are provided with a first hardened coating, and a surface contact angle of the first hardened coating is greater than or equal to 100°, so as to improve hardness and friction resistance of the surfaces. This can reduce a frictional force, so that a width of a gap between the display panel **10** and the ultra-thin glass layer **21** is close to 0, thereby improving display effect of the display module.
(58) The display module further comprises a second support assembly **60** disposed on a side of the sliding display portion **102** away from the fixed function portion **101**. The sliding display portion **102** also slides relative to the second support assembly **60**.
(59) The second support assembly **60** comprises a fixed back plate **61** close to a side of the display panel **10**, a fixed support layer **62** disposed on a side of the fixed back plate **61** away from the display panel **10**, a fixed buffer layer **63** disposed on a side of the fixed support layer **62** away from the display panel **10**, and a fixed protective glue layer **64** disposed on a side of the fixed buffer layer **63** away from the display panel **10**.
(60) A side of the display panel **10** close to the second support assembly **60** is a third sliding surface, and a side of the second support assembly **60** close to the display panel **10** is a fourth sliding surface. The third sliding surface and the fourth sliding surface are provided with a second hardened coating, and a surface contact angle of the second hardened coating is greater than or equal to 100°, so as to improve hardness and friction resistance of the surfaces. This can reduce a frictional force, so that a width of a gap between the display panel **10** and the second support assembly **60** is close to 0, thereby improving display effect of the display module.
(61) The second hardened coating is disposed on a surface of the fixed back plate **61** close to the display panel **10**.
(62) In this embodiment, a camera assembly **50** is disposed on a side of the second support assembly **60** away from the display panel **10**.
(63) In this embodiment, the polarizing layer **22** and the touch layer **24** are both disposed in the function layer **20**, and do not participate in bending and sliding of the sliding display portion **102**, thereby reducing the thickness of the sliding display portion **102**. Furthermore, this embodiment is further provided with the second support assembly **60**, and the sliding display portion **102** also slides relative to the second support assembly **60**. That is, the second support assembly **60** also does not participate in the bending and sliding of the sliding display portion **102**, which further reduces a thickness of a laminated structure participating in the bending and sliding in the display module. This improves a bendability of the sliding display portion **102**, and reduces a risk of peeling and cracking of layers of the sliding display portion **102** during the bending process. Furthermore, in this embodiment, a side of the ultra-thin glass layer **21** close to the display panel **10**, a side of the display panel **10** close to the ultra-thin glass layer **21**, a side of the display panel **10** close to the fixed back plate **61**, and a side of the fixed back plate **61** close to the display panel **10** are provided with a hardened coating, thereby improving friction resistance and hardness of sliding surfaces of the four sides and increasing a service life of the display module.
(64) Optionally, in the present disclosure, the first hardened coating, the second hardened coating, the third hardened coating, and the fourth hardened coating are all made of a same material and composition.
(65) In addition, the present disclosure further provides a display device. The display device comprises the display module described in any of the above embodiments. Please refer to FIG. **2**, FIG. **3**A, FIG. **3**B, FIG. **8**, and FIG. **9**.
(66) The display device comprises the display module and a middle frame **70**, and the display module is disposed on the middle frame **70**

(67) Specifically, the fixed function portion **101** is fixedly connected to the middle frame **70**. The sliding display portion **102** is movably connected to the middle frame **70**. The fixed function portion **101** is disposed on a light-emitting side of the sliding display portion **102**.
(68) Furthermore, the middle frame **70** is provided with a sliding assembly. The sliding assembly comprises two sliding rails **72** fixedly connected to the middle frame **70** and two sliding blocks **71** slidably connected to the sliding rails **72**. The sliding display portion **102** is fixedly connected to the sliding blocks **71**. Therefore, when the sliding blocks **71** slide on the sliding rails **72**, the sliding display portion **102** can also be driven to slide.
(69) Optionally, the sliding assembly may also comprise two sliding blocks **71** fixedly connected to the middle frame **70**, and two sliding rails **72** slidably connected to the sliding blocks **71**, The sliding display portion **102** is fixedly connected to the sliding rails **72**. Therefore, when the sliding rails **72** slides relative to the sliding blocks **71**, the sliding display portion **102** can also be driven to slide.
(70) Optionally, the sliding assembly may also comprise a track and a roller that can be movably connected, and the track and the roller are respectively disposed on the middle frame **70** and the sliding display portion **102**.
(71) The display device further comprises a driving assembly and an elastic member **78**. The driving assembly comprises an electrical machine **76**. Optionally, the elastic member **78** comprises a return spring, and the electrical machine **76** comprises a motor.
(72) One end of the elastic member **78** is connected to the middle frame **70**, and the other end of the elastic member **78** is connected to a first sub-part **1021**. The driving assembly is disposed on the middle frame **70**. One end of the driving assembly is connected to a second sub-part **1022**. When the display module needs to perform functions such as taking pictures, the second sub-part **1022** and at least part of the first sub-part **1021** are driven to bend and extend away from a side of the fixed function portion **101** by the driving assembly, so as to expose a camera assembly **50** and make the display module in a second state. At this time, the elastic member **78** is in a compressed or stretched state, and has a certain elastic potential energy. When it needs to switch to a first state, the elastic potential energy of the elastic member **78** will facilitate sliding of the sliding display portion **102** and switching to the first state.
(73) It should be noted that the driving assembly is disposed on a side of the sliding display portion **102** away from the fixed function portion **101**, and at an end of the sliding display portion **102** away from the camera assembly **50**. Therefore, the driving assembly can drive the second sub-part **1022** to slide, and drive the sliding display portion **102** to slide on the slide rails **72**. That is, the driving assembly drives at least part of the first sub-part **1021** to bend and extend to the side of the sliding display portion **102** away from the fixed function portion **101** to expose the camera assembly **50**.
(74) Furthermore, in order to maintain a sliding direction and stability of the sliding display portion **102**, the driving assembly further comprises two pulling rails **73** disposed on opposite sides of the middle frame **70**, two pulling sliders **74** slidably connected to the pulling rails **73**, and a cross bar **75** connected between the two pulling sliders **74**. The second sub-part **1022** is connected to the cross bar **75** or the pulling sliders **74**, and the electrical machine **76** is connected to the cross bar **75**. Therefore, the electrical machine **76** can pull the cross bar **75**, and the cross bar **75** slides on the pulling rails **73** via the pulling sliders **74** to drive the second sub-part **1022** to slide.
(75) In the present disclosure, the pulling rails **73**, the pulling sliders **74**, the cross bar **75**, and the electrical machine **76** are all disposed on the side of the sliding display portion **102** away from the fixed function portion **101**, and at the end of the sliding display portion **102** away from the camera assembly **50**, so that they can pull the second sub-part **1022** to slide, and drive at least part of the first sub-part **1021** to bend and extend to the side of the sliding display portion **102** away from the fixed function portion **101**, thereby exposing the camera assembly **50** and making the display module in the second state.

(76) At this time, one end of the elastic member **78** is connected to the middle frame **70**, and the other end of the elastic member **78** is connected to the sliding display portion **102**. Because the sliding display portion **102** is moved, the elastic member **78** is in the compressed or stretched state, and has a certain elastic potential energy.
(77) When the display device does not need to perform functions such as taking pictures, but needs to perform full-screen display, that is, when it needs to switch from the second state to the first state, the electrical machine **76** stops pulling the cross bar **75** or apply a reverse pushing force to a side of the cross bar **75**, which, in combination with an elastic force of the elastic member **78**, causes the sliding display portion **102** to slide in an opposite direction. Therefore, the at least part of the first sub-part **1021** bent and extended to the side of the sliding display portion **102** away from the fixed function portion **101** can be restored to the side of the sliding display portion **102** close to the fixed function portion **101** to cover the camera assembly **50** and realize full-screen display.
(78) It should be noted that the display device further comprises a power element **77** disposed on the middle frame **70** and electrically connected to the sliding display portion **102**. The power element **77** comprises a battery and a system drive board, which is not limited herein.
(79) In the above, in the present disclosure, the function layer **20** is disposed in the fixed functional portion **101** and does not slide and bend with the sliding display portion **102**, so that a thickness of a laminated structure participating in the bending in the display module is reduced, thereby reducing a strain/dislocation generated in a bending process of the sliding display portion, improving a bendability of the display module, and improving a reliability and service life of the display module.
(80) In the above embodiments, the description of each embodiment has its own emphasis. For parts not detailed in one embodiment, reference may be made to the related descriptions in other embodiments.
(81) The display substrates provided by the embodiments of the present disclosure are described in detail above. The present disclosure uses specific examples to describe principles and embodiments of the present application. The above description of the embodiments is only for helping to understand the technical solutions of the present disclosure and its core ideas. It should be understood by those skilled in the art that they can modify the technical solutions recited in the foregoing embodiments, or replace some of technical features in the foregoing embodiments with equivalents. These modifications or replacements do not cause essence of corresponding technical solutions to depart from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. A display module, comprising: a fixed function portion; and a sliding display portion disposed opposite to the fixed function portion and comprising a first sub-part close to a side of the fixed function portion and a second sub-part bent and extended from the first sub-part and away from the side of the fixed function portion; wherein the fixed function portion comprises a glass cover plate and a function layer disposed between the glass cover plate and the sliding display portion; the sliding display portion slides relative to the fixed function portion, so that the display module is switched between a first state and a second state; in the first state, the first sub-part is located on a side of the sliding display portion close to the fixed function portion; and in the second state, at least part of the first sub-part slides to a side of the sliding display portion away from the fixed function portion.
2. The display module according to claim 1, wherein the sliding display portion further comprises a display panel, a side of the display panel close to the function layer is a first sliding surface, and a side of the function layer close to the display panel is a second sliding surface.

3. The display module according to claim 2, wherein the function layer comprises a polarizing layer, and the polarizing layer is disposed on a side of the glass cover plate close to the display panel.
4. The display module according to claim 3, wherein the function layer further comprises an ultra-thin glass layer, the ultra-thin glass layer is disposed on a side of the polarizing layer close to the display panel, and a side of the ultra-thin glass layer close to the display panel is the second sliding surface.
5. The display module according to claim 4, wherein the function layer further comprises a touch layer, and the touch layer is disposed between the glass cover plate and the polarizing layer.
6. The display module according to claim 2, wherein the first sliding surface and the second sliding surface are provided with a first hardened coating, and a surface contact angle of the first hardened coating is greater than or equal to 100°.
7. The display module according to claim 2, wherein the sliding display portion further comprises a first support assembly disposed on a side of the display panel away from the fixed function portion, the first support assembly comprises a support layer, the support layer comprises a rigid part and a flexible part whose one end is connected to the rigid part and the other end is bent and away from the side of the fixed function portion, the flexible part comprises a plurality of openings, and a distribution density of the openings on a side of the flexible part close to the rigid part is less than a distribution density of the openings on a side of the flexible part away from the rigid part.
8. The display module according to claim 7, wherein the first support assembly further comprises a buffer film disposed between the support layer and the sliding display portion, and the buffer film is made of a polyurethane elastomer material.
9. The display module according to claim 1, further comprising: a second support assembly disposed on the side of the sliding display portion away from the fixed function portion, wherein the sliding display portion slides relative to the second support assembly.
10. The display module according to claim 9, wherein a side of the sliding display portion close to the second support assembly is a third sliding surface, a side of the second support assembly close to the sliding display portion is a fourth sliding surface, the third sliding surface and the fourth sliding surface are provided with a second hardened coating, and a surface contact angle of the second hardened coating is greater than or equal to 100°.
11. The display module according to claim 1, further comprising: a camera assembly disposed on the side of the sliding display portion away from the fixed function portion; wherein when the display module is in the first state, an orthographic projection of the camera assembly on the fixed function portion is located within an orthographic projection of the sliding display portion on the fixed function portion; and when the display module is in the second state, the orthographic projection of the camera assembly on the fixed function portion is outside the orthographic projection of the sliding display portion on the fixed function portion.
12. A display device, comprising a display module, wherein the display module comprises: a fixed function portion; and a sliding display portion disposed opposite to the fixed function portion and comprising a first sub-part close to a side of the fixed function portion and a second sub-part bent and extended from the first sub-part and away from the side of the fixed function portion; wherein the fixed function portion comprises a glass cover plate and a function layer disposed between the glass cover plate and the sliding display portion; the sliding display portion slides relative to the fixed function portion, so that the display module is switched between a first state and a second state; in the first state, the first sub-part is located on a side of the sliding display portion close to the fixed function portion; and in the second state, at least part of the first sub-part slides to a side of the sliding display portion away from the fixed function portion.
13. The display device according to claim 12, wherein the sliding display portion further comprises a display panel, a side of the display panel close to the function layer is a first sliding surface, a side of the function layer close to the display panel is a second sliding surface, the first sliding surface

and the second sliding surface are provided with a first hardened coating, and a surface contact angle of the first hardened coating is greater than or equal to 100°.
14. The display device according to claim 13, wherein the function layer comprises a polarizing layer, and the polarizing layer is disposed on a side of the glass cover plate close to the display panel.
15. The display device according to claim 14, wherein the function layer further comprises an ultra-thin glass layer, the ultra-thin glass layer is disposed on a side of the polarizing layer close to the display panel, and a side of the ultra-thin glass layer close to the display panel is the second sliding surface.
16. The display device according to claim 15, wherein the function layer further comprises a touch layer, and the touch layer is disposed between the glass cover plate and the polarizing layer.
17. The display device according to claim 12, wherein the display module further comprises: a camera assembly disposed on the side of the sliding display portion away from the fixed function portion; wherein when the display module is in the first state, an orthographic projection of the camera assembly on the fixed function portion is located within an orthographic projection of the sliding display portion on the fixed function portion; and when the display module is in the second state, the orthographic projection of the camera assembly on the fixed function portion is outside the orthographic projection of the sliding display portion on the fixed function portion.
18. The display device according to claim 12, further comprising a middle frame, wherein the fixed function portion is fixedly connected to the middle frame, and the sliding display portion is movably connected to the middle frame, so that the display module is switched between the first state and the second state.
19. The display device according to claim 18, wherein the middle frame is provided with a sliding assembly, the sliding assembly comprises a sliding rail fixedly connected to the middle frame and a sliding block slidably connected to the sliding rail, and the sliding display portion is fixedly connected to the sliding block.
20. The display device according to claim 18, further comprising: an elastic member, wherein one end of the elastic member is connected to the middle frame, and the other end of the elastic member is connected to the first sub-part; and a driving assembly disposed on the middle frame, wherein one end of the driving assembly is connected to the second sub-part, so that the display module is switched between the first state and the second state.